Capstone Project Submission

Instructions:

i) Please fill in all the required information.

ii) Avoid grammatical errors.

Team Member's Name, Email, and Contribution:
Name: Ravindra More **Email: ravismore1998@gmail.com** **Contribution:** • **Contributed to the project presentation and documentation.** • **Imported the dataset and required Python libraries.** • **Performed dataset understanding using head(), tail(), shape(), info(), and describe() functions.** • **Conducted data cleaning by handling missing values and removing duplicate records.** • **Analyzed null values and verified data quality.** • **Performed Exploratory Data Analysis (EDA) on customer transaction data.** • **Conducted country-wise transaction analysis.** • **Performed monthly, daily, and hourly sales trend analysis.** • **Created visualizations to understand customer purchasing behavior.** • **Performed feature engineering by extracting date and time-related features.** • **Built the Recency, Frequency, and Monetary (RFM) model.** • **Applied log transformation and feature scaling techniques.** • **Implemented K-Means Clustering for customer segmentation.** • **Used the Elbow Method to determine the optimal number of clusters.** • **Evaluated clustering quality using Silhouette Score.** • **Implemented Hierarchical Clustering and analyzed dendrograms.** • **Interpreted customer segments and generated business recommendations.** • **Extracted meaningful business insights from the analysis.** • **Contributed to conclusions, summary writing, presentation creation, and project documentation.**

Please paste the GitHub Repo link.

https://github.com/ravindramore98/Online-Retail-Customer-Segmentation

Please write a short summary of your Capstone project and its components.
Describe the problem statement, your approaches, and your conclusions. (200400 words)

The online retail business deals with thousands of customer transactions every day. Understanding customer purchasing behavior is very important because it helps businesses improve customer satisfaction, increase sales, and create better marketing strategies. Customer segmentation is a useful technique that allows businesses to group customers based on their purchasing habits and target them more effectively.

This project is based on an Online Retail dataset of a UK-based company that sells various products to customers across different countries. The dataset contains information such as Customer ID, Invoice Number, Product Description, Quantity, Unit Price, Invoice Date, and Country. The main goal of this project is to analyze customer purchasing behavior and identify different customer segments using RFM Analysis and clustering techniques.

To complete this project, I first understood the dataset and performed data cleaning by handling missing values and removing duplicate records. After cleaning the data, I carried out Exploratory Data Analysis (EDA) to identify customer purchasing patterns and business trends. Various visualizations were created to analyze transactions by country, month, weekday, and hour. Through the analysis, I found that most transactions were generated from the United Kingdom. I also observed that customer purchases were highest during October, November, and December, while the majority of transactions took place during afternoon hours.

For customer segmentation, I used Recency, Frequency, and Monetary (RFM) Analysis to measure customer value and purchasing behavior. After preparing the RFM dataset, I applied feature scaling and implemented K-Means and Hierarchical Clustering algorithms. Using the Elbow Method and Silhouette Score, the optimal number of customer segments was identified.

The project successfully grouped customers into different segments based on their purchasing behavior. These insights can help businesses identify valuable customers, improve customer retention, create personalized marketing campaigns, and make better business decisions. This project helped me understand how data analysis and machine learning techniques can be applied to solve real-world business problems and generate meaningful insights from data.